

June 19, 2026

Admissions Committee
University of Twente
Enschede, Netherlands

Dear Admissions Committee,

I am writing to express my strong interest in the PhD position focusing on Dissecting Algorithmic Collusion. As AI agents increasingly autonomously optimize pricing and market strategies, the risk of emergent, tacit collusion via reinforcement learning poses a critical regulatory and economic challenge.

As an AI & Data Science Engineer, my research has focused on understanding and controlling the emergent behaviors of complex multi-agent and dynamic systems. In my engineering theses, I modeled highly chaotic environments using advanced sequence modeling (Medical MambaTS, ST-KDFormer) and constrained them using formal mathematical boundaries (Physics-Informed Neural Networks). I possess a strong theoretical background in game theory, state-space exploration, and objective function optimization.

I am eager to apply this technical depth to dissecting the "black box" of pricing algorithms. I intend to research how deep Q-learning and multi-agent reinforcement learning algorithms arrive at collusive Nash equilibria without explicit communication. By utilizing gradient tracing, interpretability methods, and counterfactual simulations, I aim to develop robust mathematical frameworks that can detect and mitigate anti-competitive algorithmic behavior.

I look forward to discussing my technical qualifications and research proposals with you.

Sincerely,
Ismail Aissaoui
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